The Evolution of Metabolomics

At a Glance

Metabolon

Metabolomics has extraordinary utility across the life sciences, filling in gaps left by genomics and proteomics.

Despite being used for the last 50 years, it hasn't been considered a standard tool in the drug development toolbox-- until now.

1 Middle Ages

Urine properties guided physicians.

1900s-1960s -- The "Golden Age of Biochemistry" 2

Archibold Garrod postulates that human disease is caused by missing or altered steps in body's chemical pathways, establishing concept of inborn errors in metabolism (IEMs) & connection between heredity (genes), metabolic composition & phenotype.

Otto Heinrich Warburg, Gertrude Cori, Hans Adolf Krebs work to understand how biochemistry impacts complex biological processes. Through their studies, first metabolic pathway maps were created. Today we use this framework to derive mechanistic understanding from the measurement of metabolites.

3 Late 20th Century

- Crude urine quality (bioassays, glucose)
- Clinical chemistry (bilirubin, creatinine, triglycerides, cholesterol)
- Inborn errors and Specialty testing (acyl-carnitines)
- Laboratory testing and diagnostics (Vitamin D, Steroids)

LAB TESTED

21st Century and Beyond 4

In recent years, metabolomics has contributed to a number of critical discoveries to advance drug development and approval.

The phenotyping power of metabolomics makes it a pivotal tool for understanding living systems and has a wide variety of applications, including:

- Identify targets for drug development
- Elucidate mechanism of action
- Translate pre-clinical models for superior clinical trial design
- Evaluate PK/PD for drug dosing
- Stratify patients by identifying responders/non-responders
- Monitor drug toxicity
- Bioprocess optimization

Metabolon

ABOUT METABOLON

Metabolon, Inc., is the global leader in revealing new biological insights through metabolomics, harnessing the power of its Precision Metabolomics™ Platform to provide biological insights of disease state and physiological reactions in the present time. The company offers scalable, customizable metabolomics solutions from discovery through clinical trials and product life-cycle management. Leveraging one of the world's most diverse and rich patient data sets, Metabolon is equipped to deliver biologically relevant insights to address some of the most difficult and pressing questions in the life sciences, helping to accelerate research and product development success in the biopharma, population health, consumer products, wellness and academic research sectors. For more information, please visit www.metabolon.com or follow us on LinkedIn or Twitter.

The Rise
of Metabolomics

Powering Discoveries One Metabolite At a Time

As a unique, real-time fingerprint of biological systems, Metabolomics is no stranger to the scientific community.

Yet, this youngest member of the "omics group, has recently experienced a rapid increase in adoption thanks to its ability to provide actionable information that can advance research, clinical development, and approvals.

The proof is in the data:

- **1.7M+ publications on metabolites and lipids.[1]**

And it's not just for basic research.

- **Over 18k clinical trials have been using metabolites or lipids as endpoints, of these over 7,000 studies were funded by industry alone.[2]**

Growing use of Metabolites & Lipids in Clinical Trials

18,000+ Clinical Trials Using Metabolites Or Lipids As Endpoints

1 https://pubmed.ncbi.nlm.nih.gov/
2 https://www.clinicaltrials.gov/

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Still, there are challenges to overcome.

- Labs rarely have comprehensive metabolomics technologies nor are equipped to handle accurate unbiased metabolite identification.
- Interpreting the vast amount of data is slow, if not impossible.
- Researchers lack access to an extensive library of metabolites.

Imagine if there was a way to:

- Develop a systems biology understanding of human response
- Gain functional insights on how DNA and proteins interact with diet and environment
- Deliver on the promise of personalized medicine with deep phenotypic knowledge

We have the solution.

With over 20 years of experience deciphering thousands of discrete chemical signals from genetic and non-genetic factors to discover biomarkers and reveal biological pathways, Metabolon provides actionable biological insights to answer the toughest questions.

Metabolon
Sets the Standard

5,200+

METABOLITES

World's largest **reference library**

comprehensive coverage across over 70 metabolic pathways

ISO 9001:2015

certified for analytical and diagnostic testing

Certified by CLIA and accredited by the College of American Pathologists for diagnostic testing on human samples

2,000+

PUBLICATIONS

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